

Tristan Brideweser

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Education

Purdue University - West Lafayette, IN

- MSE, Master of Science in **Interdisciplinary Engineering/Robotics** (Exp. May 2027)
- MBT, Master of **Business & Technology** (Exp. May 2027)
- BS, Bachelor of Science in **Data Science, Applied Statistics**; Minor: **Design & Innovation** (May 2025)

Relevant Coursework: Artificial Intelligence, Autonomous Systems, Data Mining & Machine Learning, Accounting for Managers, DevOps & Technology Solutions

Work Experience

- **Project Lead** | The Data Mine Corporate Partners Program - Purdue University | Dec 2022 - Present
 - **Johnson & Johnson Innovative Medicine:** Applying GenAI methods to identify subgroups within biomedical datasets, improving model interpretability for clinical insights.
 - **CTB, Inc.:** Built predictive maintenance models for poultry houses, achieving >85% anomaly detection accuracy and reducing downtime risk.
 - **Knudsen Institute:** Designed ARIMA + DTW clustering dashboards to analyze procurement trends in the automotive supply chain to forecast surge events 2 weeks earlier.
 - **MISO:** Developed regression models with >80% accuracy to forecast transmission line ratings, supporting compliance with FERC Order 881.
- **Teaching Assistant** | Mitchell E. Daniels School of Business - Purdue University | Feb 2024 - May 2024
 - Designed a Python–Qualtrics survey tool to assess student development in core leadership skills regarding client relationships, used by 200+ students per semester.
- **Admissions Representative** | Office of Admissions - Purdue University | Sept 2023 - Present
 - Delivering presentations to 10,000+ students/families annually, improving Purdue's admission yield.
- **Artificial Intelligence Research Intern** | Weslabs.ai - Boston, MA (Remote) | May 2023 - Sept 2023
 - Assembled database infrastructure for AI-driven educational applications and performed market research to identify potential clients.

Research & Projects

- **Undergraduate Researcher** | Purdue UAS Research and Testing (PURT) | Aug 2024 - Dec 2024
 - Improved autopilot accuracy of UAS platforms using Python-based Kalman filtering to reduce flight time by 10%.
- **Algorithms Team Lead & Outreach Coordinator** | Autonomous Robotics Club of Purdue (ARC) | Jan 2023 - May 2025
 - Directed algorithm research subteam of 10 computer science and engineering students developing Kalman filters and Viterbi algorithms, enabling a robotic hand to perform piano pieces.
 - Coordinated partnerships with faculty & industry sponsors, securing funding and resources.
- **Co-Founder & Data Scientist** | Startup Project - Spoke & Wing, LLC | Jan 2022 - May 2023
 - Developed predictive model estimating ~\$25–30M lifetime savings and patenting aviation fuel tank regulator to minimize leakage rates.
 - 1st Place in Design & Innovation Challenge 2023; 2nd Place in Moonshot Pitch Challenge 2024.

Leadership & Service

- **Mentor** | Purdue FIRST Program | Aug 2025 - Present
 - Assisting K-12 robotics teams in design, coding, and strategy.
- **Executive Ambassador** | The Data Mine - Purdue University | Sept 2023 - Jan 2025
 - Collaborated with corporate partners, faculty, and prospective students while enhancing student development opportunities.
- **Eagle Scout** | Boy Scouts of America | Awarded June 2021
 - Demonstrated leadership, project planning, and community service.

Skills

Technical: Python/R, Git, ROS2, Fanuc/Yamaha Robotics, Fusion 360, EAGLE

Business/Leadership: Agile Project Management, Team Leadership, Strategic Communication